



DAFFODIL INTERNATIONAL UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
FYDP (PHASE-I) EVALUATION REPORT
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Project Identification:

I. Project Title	A Unified Explainable and Robust Deep Learning Framework for Multimodal Brain Tumor MRI Segmentation	
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V. Submission Date:	03 August 2026	
VI. Certificate:	“This is to certify that the final year design project work until Phase-I evaluation held on 03 August 2026, titled as stated in <i>Sec. I</i> , executed by the students’ group mentioned in <i>Sec. II</i> , have been found satisfactory and every section of this report is reflecting the same.”	(Signature of Supervisor & date)

Project Insights

Thematic Area(s): <i>[Just click the check box]</i>	Artificial Intelligence and Machine Learning	☒
	Deep Learning	☒
	Health Informatics	☒
	Cybersecurity	☐
	Software Engineering and Development	☐
	Blockchain Technology	☐
	Internet of Things (IoT)	☐
	Computer Networks	☐
	Computer Vision	☒
	Natural Language Processing (NLP)	☐
	Robotics	☐
	Game Development	☐

	Cloud Computing	☐
	Image Processing	☒
	Others (please specify):	
Software packages, tools, and programming languages	Programming Language: Python Software Packages: PyTorch, NumPy, Pandas, OpenCV, NiBabel, Matplotlib, h5py, segmentation-models-pytorch (SMP), torchvision Development Tools: Google Colab (Pro), Google Drive Hardware Platform: NVIDIA CUDA-enabled GPU (Google Colab T4), Automatic Mixed Precision (AMP)	

CO Description for FYDP-Phase-I

CO	CO Descriptions	PO
CO4	Perform economic evaluation, cost estimation, and apply appropriate project management practices throughout the FYDP lifecycle in the development of an Explainable and Robust AI-Based Brain Tumor MRI Segmentation Framework.	PO1 1
CO6	Select and apply appropriate deep learning methodologies, computational resources, and contemporary engineering and IT tools for modeling, segmentation, and solving complex problems related to multimodal brain tumor MRI segmentation.	PO5
CO7	Assess societal, health, safety, legal, and ethical responsibilities associated with the development and application of AI-driven brain tumor MRI segmentation systems in healthcare environments.	PO6
CO10	Operate effectively as an individual and as a member of a multidisciplinary team during the FYDP while collaboratively developing and evaluating an AI-based brain tumor MRI segmentation framework.	PO9

1. Project Overview

1.1 Introduction

Brain tumors are among the most challenging neurological disorders and remain a major cause of morbidity and mortality worldwide. Early diagnosis and accurate assessment are essential for effective treatment planning, surgical intervention, radiotherapy, and continuous disease monitoring. Magnetic Resonance Imaging (MRI) is the preferred imaging modality for brain tumor diagnosis because it provides high-resolution visualization of brain tissues without exposing patients to ionizing radiation [6].

Multimodal MRI, including T1-weighted (T1), contrast-enhanced T1-weighted (T1ce), T2-weighted (T2), and Fluid-Attenuated Inversion Recovery (FLAIR) sequences, provides complementary anatomical and pathological information that enables comprehensive visualization of tumor regions, surrounding edema, and enhancing tissues [8]. Accurate segmentation of these tumor regions is a fundamental step in clinical practice because it assists clinicians in diagnosis, treatment planning, radiotherapy target delineation, and treatment response evaluation [15].

Traditionally, brain tumor segmentation is performed manually by experienced radiologists. However, manual delineation is labor-intensive, time-consuming, and susceptible to inter-observer variability, making it difficult to maintain consistent segmentation quality in routine clinical practice [4]. Consequently, the development of automated brain tumor segmentation techniques has become an active research area aimed at improving efficiency, accuracy, and reproducibility.

Recent advances in deep learning, particularly convolutional neural network (CNN)-based architectures, have significantly improved the performance of automated brain tumor segmentation systems. Encoder-decoder architectures such as U-Net and its variants have demonstrated remarkable capability in learning hierarchical image representations directly from multimodal MRI data [1]. More recently, attention mechanisms and Vision Transformer-based architectures have further enhanced segmentation performance by effectively capturing both local image features and long-range contextual relationships [23].

The availability of publicly accessible benchmark datasets has also accelerated research in this field. Among them, the Brain Tumor Segmentation (BraTS) challenge has become the most widely adopted benchmark for developing and evaluating brain tumor segmentation algorithms by providing standardized multimodal MRI datasets and evaluation protocols [2]. Continuous improvements in the BraTS challenges have further encouraged the development of more accurate and clinically relevant segmentation methods [5].

Despite these advancements, existing studies primarily focus on improving segmentation accuracy, while comparatively less attention has been given to the explainability and robustness of deep learning models [13]. Since deep learning systems are increasingly being considered for clinical decision support, it is essential not only to achieve high segmentation performance but

also to understand the reasoning behind model predictions and evaluate their reliability under different imaging conditions [12].

Motivated by these challenges, this study proposes "A Unified Explainable and Robust Deep Learning Framework for Multimodal Brain Tumor MRI Segmentation." The proposed framework aims to develop a comprehensive deep learning pipeline that combines accurate brain tumor segmentation with explainability and robustness evaluation within a unified experimental framework. Using the publicly available BraTS 2021 dataset, the proposed research will systematically perform data preprocessing, patient-level data partitioning, model development, performance evaluation, explainability analysis, and robustness assessment to develop a more transparent, reliable, and clinically applicable brain tumor MRI segmentation system.

1.2 Background Study

Brain tumor segmentation has progressed from conventional image-processing methods to deep learning-based systems. Earlier approaches used thresholding, clustering, handcrafted features, and machine-learning classifiers, but these methods were often sensitive to noise, intensity variation, and irregular tumor boundaries [19].

CNN-based models reduced the need for manual feature engineering by learning hierarchical representations directly from MRI data. U-Net and its variants became widely used because their encoder-decoder structure and skip connections support accurate spatial reconstruction [17]. Three-dimensional models further preserve inter-slice context, although they require greater memory and computation [11].

Recent studies have introduced residual learning, attention mechanisms, multimodal feature fusion, and deep supervision to improve segmentation performance. AD-Net, for example, was designed to reduce interference among MRI modalities and improve whole tumor, tumor core, and enhancing tumor segmentation [10]. Hybrid CNN-Transformer models have also gained attention because they combine local feature extraction with long-range contextual modeling [9].

The BraTS challenges have provided standardized multimodal MRI datasets and evaluation protocols, supporting consistent comparison among segmentation methods [18]. However, recent reviews indicate that many studies still focus mainly on segmentation accuracy, while explainability, robustness, computational efficiency, and failure analysis remain comparatively underexplored [8].

External validation also remains limited. Boyd et al. showed that a model trained on adult glioma data required domain-specific transfer learning to perform effectively on pediatric low-grade glioma MRI [21]. This finding highlights the importance of evaluating model reliability under different populations and imaging conditions.

Overall, existing CNN-, U-Net-, attention-, and Transformer-based methods have achieved substantial progress in brain tumor MRI segmentation. Nevertheless, a unified framework is still needed to jointly evaluate segmentation performance, explainability, and robustness under a consistent experimental protocol.

Table 1.1. Comparison of Recent Brain MRI Segmentation Papers

Study	Year	Method	Dataset	Main Finding	Limitation
Nizamani et al. [1]	2023	Feature-enhanced hybrid U-Net	Public MRI datasets	Improved segmentation through feature enhancement and fusion	Complex preprocessing; no XAI evaluation
Peng and Sun [10]	2023	AD-Net	BraTS 2020	Effective multimodal feature separation and subregion segmentation	Lower performance for smaller tumor regions
Guan et al. [11]	2022	3D AGSE-VNet	BraTS 2020	Combined channel attention with 3D segmentation	High computational demand
Ottom et al. [17]	2022	Znet	LGG MRI dataset	Strong 2D tumor segmentation using encoder–decoder learning	Binary segmentation only
Sangui et al. [20]	2023	Modified 3D U-Net	BraTS 2020	Demonstrated the effectiveness of volumetric U-Net segmentation	Limited region-wise reporting
Angona and Mondal [9]	2025	Residual Attention U-Net with Swin Transformer	BraTS 2019/2020	Combined local and global contextual modeling	Increased model complexity
Boyd et al. [21]	2024	nnU-Net with stepwise transfer learning	BraTS 2021 and pediatric cohorts	Improved external pediatric segmentation through domain adaptation	Single-sequence pediatric segmentation
Alqhtani et al. [14]	2024	FCM-SVM framework	CE-MRI	Effective conventional segmentation and classification pipeline	Not directly comparable with multiclass BraTS segmentation

1.3 Gap Analysis

Although significant progress has been achieved in brain tumor MRI segmentation through CNNs, U-Net variants, attention mechanisms, and Transformer-based architectures, several research gaps still remain. Most existing studies primarily emphasize segmentation accuracy and Dice score while providing limited investigation into the reliability and interpretability of model predictions [8].

Many recent segmentation frameworks incorporate attention mechanisms or Transformer modules to improve feature learning; however, they rarely perform systematic explainability analysis to verify whether the highlighted regions truly correspond to clinically meaningful tumor structures [23]. As a result, the decision-making process of these models remains largely a black box.

Another important limitation is the lack of robustness evaluation. Most studies report performance only on clean benchmark datasets without investigating how segmentation performance changes under image noise, intensity variation, blur, or other clinically relevant perturbations [13]. Consequently, the real-world reliability of these models remains uncertain.

Furthermore, several high-performing architectures are computationally expensive because they rely on complex 3D networks or hybrid CNN–Transformer designs, making them difficult to deploy in resource-constrained environments [9]. Many studies also focus mainly on segmentation performance while giving comparatively less attention to computational efficiency and reproducible experimental pipelines [18].

Based on the identified gaps, this research proposes a unified framework that integrates accurate multimodal brain tumor MRI segmentation with explainability and robustness evaluation within a single experimental pipeline. The proposed study also emphasizes a reproducible implementation strategy to facilitate reliable model development, evaluation, and future clinical applicability.

2. Objectives

The primary objective of this research is to develop a reliable and reproducible deep learning framework for multimodal brain tumor MRI segmentation. In addition to achieving accurate tumor segmentation, the proposed framework aims to enhance model transparency and reliability by incorporating explainability and robustness evaluation. The specific objectives of this study are as follows:

- To develop a unified deep learning framework for accurate multimodal brain tumor MRI segmentation using the BraTS 2021 dataset.
- To integrate Explainable Artificial Intelligence (XAI) techniques for interpreting and improving the transparency of segmentation model predictions.

- To evaluate the robustness and reliability of the developed models under different image perturbations using comprehensive quantitative performance metrics.
- To compare representative CNN-based and Transformer-based segmentation models in terms of segmentation accuracy, explainability, robustness, and computational efficiency under a unified experimental framework.

3. Methodology:

3.1 Research Design

This study adopts an experimental and quantitative research design to develop and evaluate a unified deep learning framework for multimodal brain tumor MRI segmentation. The proposed methodology follows a systematic and reproducible workflow, beginning with dataset acquisition and preprocessing, followed by patient-level data partitioning, model development, performance evaluation, explainability analysis, and robustness assessment. A patient-level data splitting strategy is employed to prevent data leakage and ensure fair model evaluation throughout the experimental process.

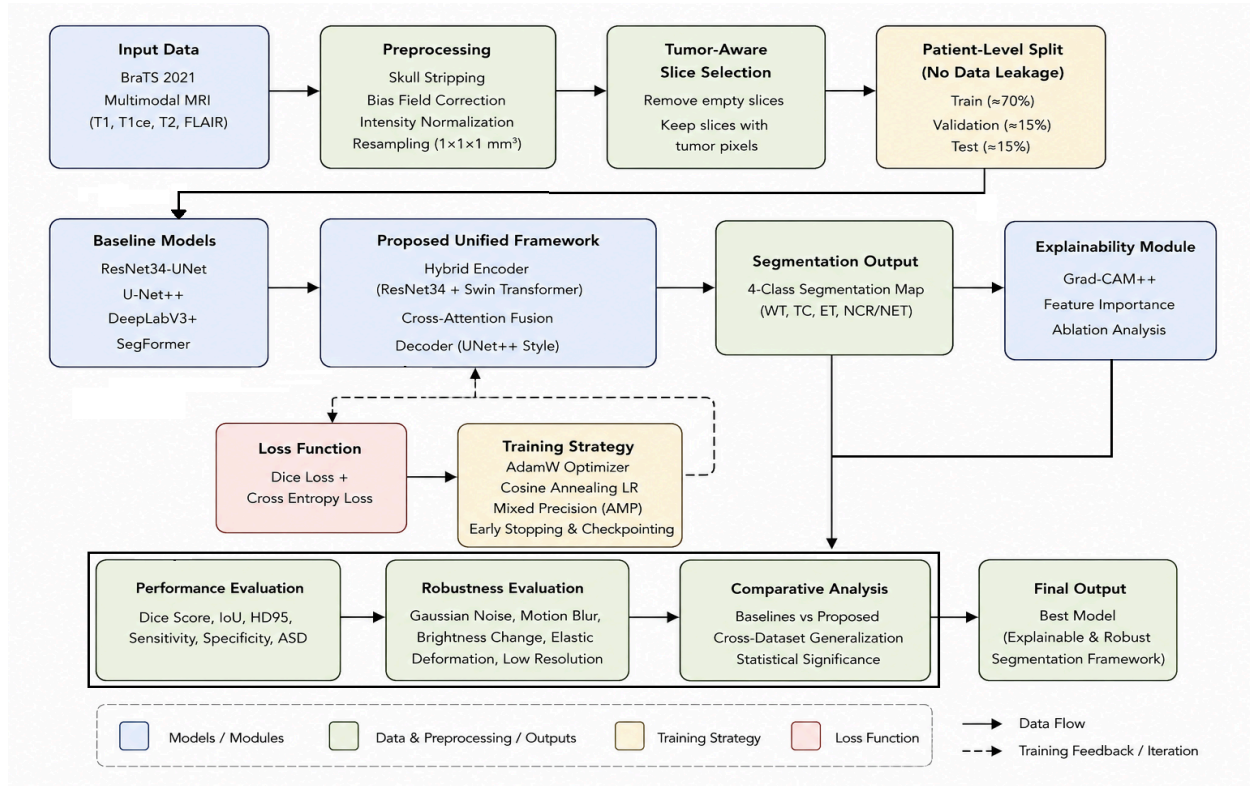


Figure 3.1. Overall Research Workflow

The research utilizes the publicly available BraTS 2021 multimodal brain MRI dataset, consisting of T1, T1ce, T2, and FLAIR modalities with expert-annotated tumor masks. Representative CNN-based and Transformer-based segmentation models will be implemented

under a unified experimental protocol to ensure a fair comparison. The developed models will be evaluated using standard segmentation metrics, while Explainable Artificial Intelligence (XAI) techniques and robustness analysis will be incorporated to assess the transparency and reliability of model predictions.

The overall research workflow adopted in this study is illustrated in Figure 1.

3.2 Data Collection

The data used in this study were collected from the publicly available Brain Tumor Segmentation (BraTS) 2021 dataset, one of the most widely recognized benchmark datasets for brain tumor MRI segmentation research. The dataset consists of expert-annotated multimodal magnetic resonance imaging (MRI) scans collected from multiple institutions under standardized acquisition and annotation protocols [2].

Each patient case contains four co-registered MRI modalities, namely T1-weighted (T1), contrast-enhanced T1-weighted (T1ce), T2-weighted (T2), and Fluid-Attenuated Inversion Recovery (FLAIR), along with the corresponding ground-truth tumor segmentation mask. The original segmentation labels include background (0), necrotic/non-enhancing tumor core (1), peritumoral edema (2), and enhancing tumor (4). During preprocessing, the labels are remapped into contiguous classes to facilitate multiclass deep learning segmentation.

For this research, the complete BraTS 2021 Training Dataset was utilized. After archive verification and integrity checking, a total of 1,251 complete patient cases were identified, with no missing MRI modalities or segmentation masks. The dataset was initially stored in Google Drive and subsequently extracted into the Google Colab runtime environment to improve data loading speed and training efficiency. All preprocessing, dataset validation, patient-level splitting, and subsequent experiments were performed using this validated dataset to ensure reproducibility throughout the study [18].

3.3 Analysis Techniques

To ensure a systematic and reproducible experimental process, this study follows a unified analysis pipeline covering data preparation, model development, evaluation, explainability analysis, and robustness assessment. Initially, the BraTS 2021 multimodal MRI dataset undergoes preprocessing, including data integrity verification, patient-level data partitioning, intensity normalization, label remapping, image resizing, and tumor-aware slice selection. A leakage-free patient-level train-validation-test split is employed to ensure unbiased model evaluation.

Representative CNN-based and Transformer-based segmentation models are then implemented under a common experimental framework. All models are trained using identical preprocessing procedures, optimization strategies, and evaluation protocols to enable fair comparison. Performance is assessed using widely adopted segmentation metrics, including Dice Score,

Intersection over Union (IoU), Precision, Recall, Specificity, and the 95th Percentile Hausdorff Distance (HD95).

To improve model transparency, Explainable Artificial Intelligence (XAI) techniques will be incorporated to visualize and interpret the segmentation decisions of the trained models. Furthermore, robustness analysis will be conducted by evaluating model performance under various image perturbations to investigate prediction stability and reliability. Finally, the developed models will be compared based on segmentation accuracy, explainability, robustness, and computational efficiency to identify the most effective framework for multimodal brain tumor MRI segmentation.

4. Progress Achieved:

4.1 Completed Tasks

The study commenced with an extensive review of recent literature to understand the current state of brain tumor MRI segmentation, identify existing limitations, and establish the research gap. Based on the findings, the research objectives, overall methodology, and experimental workflow were designed for the proposed unified explainable and robust deep learning framework.

The publicly available BraTS 2021 multimodal brain MRI dataset was then collected and thoroughly validated through archive verification, runtime extraction, data integrity checking, and patient-level quality assessment. All patient cases were examined to ensure the availability of the required MRI modalities and corresponding ground-truth segmentation masks before proceeding to model development.

Following dataset validation, a patient-level train-validation-test partitioning strategy was implemented to eliminate data leakage during model evaluation. Tumor-aware slice selection was subsequently performed to construct an efficient training dataset while preserving complete validation and testing datasets for unbiased performance assessment.

A comprehensive preprocessing framework was developed to prepare the multimodal MRI data for deep learning. The framework included image normalization, label remapping, image resizing, quality verification, and preprocessing validation to ensure consistency throughout the experimental pipeline. A PyTorch-based data pipeline was then implemented together with an HDF5 runtime caching mechanism to improve data loading efficiency and accelerate model training.

Finally, a universal deep learning training engine was developed with support for automatic checkpointing, resume functionality, mixed-precision training, early stopping, and experiment logging to ensure reproducibility and efficient experimentation. The implementation of the baseline brain tumor segmentation model was also initiated, providing the foundation for subsequent model training, comparative analysis, explainability evaluation, and robustness assessment.

4.2 Results Obtained

The implemented experimental pipeline successfully established a complete and reproducible deep learning environment for multimodal brain tumor MRI segmentation. The BraTS 2021 dataset was successfully validated, preprocessed, and partitioned using a patient-level splitting strategy, ensuring leakage-free training, validation, and testing. The preprocessing framework, PyTorch data pipeline, runtime HDF5 caching mechanism, and universal resume-supported training engine were also successfully developed and validated.

The first baseline segmentation model, ResNet34-UNet, was successfully trained using the proposed training framework. The model completed training with an early stopping strategy and achieved its best validation performance at Epoch 19, while training terminated at Epoch 29. The highest validation Dice Score obtained by the baseline model was 0.9211, demonstrating the effectiveness of the developed preprocessing pipeline, training engine, and optimization strategy. Furthermore, the training process successfully utilized automatic checkpointing, mixed-precision training, resume functionality, and experiment logging, confirming the stability and reproducibility of the proposed implementation framework.

The obtained results establish a strong baseline for subsequent implementation and comparison of additional segmentation architectures, including U-Net++, DeepLabV3+, SegFormer, and the proposed unified explainable and robust deep learning framework.

5. Challenges Faced:

S.No.	Issues and Challenges	Strategies or Plans
1	Large-scale BraTS 2021 dataset required significant storage and loading time.	The dataset was stored in Google Drive and extracted to the Google Colab runtime to improve I/O performance during training.
2	Frequent Google Colab runtime disconnections interrupted long-running experiments.	A resume-supported training framework with automatic checkpointing and experiment logging was implemented to continue training from the last saved state.
3	Loading NIfTI (.nii.gz) files during every training iteration significantly reduced training speed.	An HDF5-based runtime caching mechanism was developed to accelerate data loading and improve overall training efficiency.

4	Preventing data leakage between training and evaluation datasets was essential for fair model assessment.	A patient-level train–validation–test partitioning strategy was adopted to ensure complete separation of patient data across all experimental sets.
5	Severe class imbalance between tumor and background regions could bias model learning.	Tumor-aware slice selection was performed to increase the representation of informative tumor slices during training while maintaining unbiased evaluation datasets.
6	Processing multimodal MRI data required consistent preprocessing across all modalities.	A unified preprocessing pipeline was developed, including normalization, label remapping, image resizing, and data validation to ensure consistency throughout the experiments.
7	The high computational requirements of deep learning models posed challenges for efficient experimentation.	GPU acceleration, mixed-precision training (AMP), optimized PyTorch data loading, and runtime caching were utilized to improve computational efficiency.

6. Next Steps:

S.No.	Next Task	Estimated Completion Time (MM-YY)
1	Completion of baseline model benchmarking (U-Net, U-Net++, DeepLabV3+, SegFormer, and other selected architectures)	09–26
2	Development and implementation of the proposed unified explainable and robust deep learning framework	09–26
3	Integration of Explainable Artificial Intelligence (XAI) techniques and comprehensive explainability analysis	10–26

4	Robustness evaluation under various image perturbations, sensitivity analysis, and comparative performance evaluation	10–26
5	Comprehensive experimental analysis, discussion, and validation of the proposed framework	11–26
6	Final thesis writing, result refinement, defense preparation, and manuscript preparation	12–26

7. Updated Timeline:

Tasks	Weeks																	
	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Literature Review	■	■																
Dataset Collection					■	■												
Data Preprocessing							■	■										
Pipeline Development								■	■									
Methodology Design										■	■							
Model Training Setup												■	■	■				
Baseline Model Running															■	■	■	■

Estimated Work Period	■
Actual Work Period	■

8. Resources Utilized:

This phase of the project utilized the publicly available BraTS 2021 multimodal brain MRI dataset, cloud-based computational resources, and open-source deep learning frameworks. Model development and experimentation were conducted using Python and PyTorch within Google Colab Pro, leveraging NVIDIA Tesla T4 GPU acceleration for efficient model training. Medical

image processing and preprocessing were performed using NiBabel, OpenCV, NumPy, and Pandas, while data visualization was carried out using Matplotlib. To improve training efficiency, an HDF5-based runtime caching mechanism was implemented, and all datasets, model checkpoints, experiment logs, and generated artifacts were managed through Google Drive. The overall implementation emphasizes reproducibility, computational efficiency, and a resume-supported experimental workflow.

9. Project Management and Financial Analysis:

The successful execution of this research required both computational resources and operational expenses throughout the project. The primary expenditure was associated with cloud-based GPU computing using Google Colab Pro, which facilitated efficient deep learning model training and experimentation. Additional resources, including internet connectivity, cloud storage, electricity, and documentation expenses, were utilized to support dataset management, experimental development, and report preparation. Table X presents the estimated expenditure incurred during the current phase of the project.

S.No	Item	Description	Duration	Cost (BDT)
1	Google Colab Pro	Cloud GPU (NVIDIA Tesla T4) for deep learning model training and experimentation	2 Months	3,600
2	Internet Subscription	Broadband internet for literature review, dataset download, cloud computing, and experiment synchronization	2 Months	2,400
3	Google Drive Storage	Cloud storage for BraTS 2021 dataset, HDF5 cache, checkpoints, logs, and experimental artifacts	2 Months	600
4	Electricity	Power consumption for long-duration computing, data preparation, and development activities	2 Months	300
5	Report Printing	Printing of progress report and supporting documents	One-time	250

6	Miscellaneous	Stationery, contingency expenses, and other research-related costs	—	850
	Total			8,000

10.Future Considerations:

The current work establishes a strong foundation for developing an explainable and robust deep learning framework for multimodal brain tumor MRI segmentation. In the subsequent stages, additional baseline models will be implemented and benchmarked under a unified experimental protocol. The proposed framework will then be developed by integrating Explainable Artificial Intelligence (XAI) techniques to improve the interpretability of model predictions. Furthermore, comprehensive robustness evaluation will be conducted under various image perturbations to assess model reliability and generalization. Finally, comparative performance analysis, experimental validation, and detailed documentation will be completed to establish a clinically reliable and reproducible brain tumor MRI segmentation framework.

11.Conclusion:

This report presented the progress achieved in developing a unified explainable and robust deep learning framework for multimodal brain tumor MRI segmentation. The research successfully established a reproducible experimental environment, including dataset validation, preprocessing, patient-level data partitioning, data pipeline development, runtime optimization, and a resume-supported training framework. Furthermore, the baseline model development was successfully initiated, providing the foundation for subsequent comparative experiments.

The completed work demonstrates that the proposed research is progressing according to the planned methodology and has established the essential infrastructure required for the remaining stages of the project. The forthcoming work will focus on implementing additional segmentation models, integrating explainability techniques, performing robustness evaluation, and conducting comprehensive comparative analysis to achieve the overall research objectives. The outcomes of this research are expected to contribute toward the development of more accurate, transparent, and reliable deep learning systems for multimodal brain tumor MRI segmentation.

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Appendix

Appendix A: BraTS 2021 Dataset and Experimental Splits

This study utilizes the publicly available BraTS 2021 multimodal brain MRI dataset to develop and evaluate the proposed explainable and robust deep learning framework. The dataset consists of four MRI modalities (T1, T1ce, T2, and FLAIR) with expert-annotated multiclass tumor segmentation masks. After dataset verification and integrity checking, patient-level data splitting was performed to eliminate data leakage between training, validation, and testing sets.

Table A1: Summary of dataset and experimental splits

Dataset	Total Patients	Train	Validation	Test	Usage
BraTS 2021	1,251	875	188	188	Training, validation, and performance evaluation

N.B. Patient-level splitting was adopted to prevent data leakage. Tumor-aware slice selection was applied to improve training efficiency, while complete patient volumes were preserved for validation and testing.

Appendix B: Baseline Segmentation Models

Several representative deep learning segmentation architectures were selected as baseline models to provide a fair and comprehensive comparison with the proposed framework.

Table B1: Baseline models used in this study

Model	Architecture Type	Key Characteristics
ResNet34-UNet	CNN Encoder–Decoder	Residual feature extraction with U-Net decoder
U-Net++	Nested CNN	Dense skip connections for feature refinement
DeepLabV3+	CNN with Atrous Convolution	Multi-scale contextual feature extraction
SegFormer	Transformer-Based	Lightweight transformer with global contextual modeling

Proposed Model	Hybrid CNN–Transformer	Explainability-guided and robustness-aware segmentation framework
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Appendix C: Training Configuration and Computational Resources

All experiments were conducted under a unified training protocol to ensure reproducibility and fair comparison among all baseline and proposed models.

Table C1: Training configuration

Parameter	Value
Dataset	BraTS 2021
Input Size	192×192
Input Channels	4 (T1, T1ce, T2, FLAIR)
Output Classes	4
Batch Size	8
Optimizer	AdamW
Initial Learning Rate	3×10^{-4}
Loss Function	Dice Loss + Cross-Entropy Loss
Mixed Precision	Enabled (AMP)
Early Stopping	Enabled
Runtime Cache	HDF5
Platform	Google Colab Pro

Hardware	NVIDIA Tesla T4 GPU
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N.B. All baseline models were trained using the same preprocessing pipeline, optimization strategy, patient-level data splitting protocol, and evaluation framework to ensure fair and reproducible performance comparison.